

Multiple Choice

1. **Answer:** E

Options: A) The heart synthesizes its own nutrients internally without reliance on external sources.; B) The heart relies on neural control for its metabolic needs.; C) The heart extracts nutrients from the atria and ventricles.; D) The heart's metabolic needs are met by the lymphatic system.; E) The heart depends on its own blood supply from the coronary vessels.

Note: Correct option: E - The heart depends on its own blood supply from the coronary vessels.

2. **Answer:** B

Options: A) a large population; B) genetic drift; C) random mating; D) absence of selection

Note: Correct option: B - genetic drift

3. **Answer:** A

Options: A) The drop in blood pressure causes blood clotting; B) The drop in blood pressure causes an increase in plasma protein concentration; C) The drop in blood pressure stimulates the excretion of fluids by the kidneys; D) The drop in blood pressure causes dehydration; E) The drop in blood pressure triggers the release of antidiuretic hormone

Note: Correct option: A - The drop in blood pressure causes blood clotting

4. **Answer:** D

Options: A) Coevolution; B) Punctuated equilibrium; C) Directional selection; D) Divergent evolution; E) Convergent evolution

Note: Correct option: D - Divergent evolution

5. **Answer:** A

Options: A) positive chemotropism; B) positive phototropism; C) negative phototropism; D) negative chemotropism; E) positive hydrotropism

Note: Correct option: A - positive chemotropism

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'robin, dog'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
10. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** Ontogeny refers to the development of social behaviors in humans from infancy to adulthood.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** The vesicle remains intact and floats freely in the cytoplasm.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key